

Prime counts and prime sums from multiplication tables: a self-similar formula system

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Abstract

A multiplication table whose entries are all at least 2 manufactures exactly the composite numbers; the primes are precisely what the table cannot make. Taking this observation seriously as a computational program — “sum of the integers, minus the de-duplicated output of the table, equals the sum of the primes” — forces a de-duplication problem, since a composite m appears $d(m) - 2$ times in the table, and $2^{\Omega(m)}$ -fold growth in higher-dimensional tables. Its exact resolution is the alternating-harmonic combination of k -dimensional multiplication tables, which is Linnik’s identity. We develop this into a three-line inductive formula family S_j computing $\sum_{p \leq y} p^j$ for every weight $j \geq 0$ — counts at $j = 0$, sums at $j = 1$, power sums beyond — over arbitrary intervals, with a self-similar “dust” recursion that terminates in *empty* tables, so that no prime, primality test, or sieve appears anywhere in the system, at any depth. We prove the interval structure: sums live on hyperbola shells, folded coordinates are root-bounded, evaluation requires only the finite quotient–root lattice of the endpoint, and dimension k is *born* at 2^k with a single newborn. We prove a duplicated-mass theorem: the 2-dimensional table cancels identically out of its own duplication ledger, which is settled by the diagonal and the tables of dimension ≥ 3 alone. A reference implementation passes forty exact interval tests for counts and sums, and at the billion scale reproduces $\pi(10^9) = 50,847,534$ together with the primality and index of 999,999,937 in minutes, touching 63,719 lattice points out of 10^9 integers. A curious observable falls out: stepping the endpoint from y to $y + 1$ enlarges the evaluation lattice exactly at the divisors of $y + 1$, so a prime step adds a single point — primality is visible as minimal lattice growth. The core identity is Linnik’s, the resulting algorithms sit in the Legendre–Meissel family, and the computational route through Linnik’s identity was pioneered by McKenzie; the contribution is the geometric derivation, the duplicated-mass and layer-birth theorems, the interval taxonomy, and the verification discipline.

Keywords: prime counting function; prime power sums; Linnik’s identity; multiplication tables; divisor function; Legendre–Meissel methods; machine-verified computation.

MSC 2020: primary 11N05; secondary 11A25, 11Y35, 11Y16.

*Version 1.0, August 2026. Companion to *Certified exact sums of primes over dyadic intervals via a ladder of generalized factorials* [22] (same author). Developed in collaboration with Claude (Anthropic); all numerical claims were machine-verified as described in Section 6. Responsibility for the content rests with the author. A full disclosure of AI use appears at the end of the paper.

1 Introduction

Let $U_k(N)$ denote the k -dimensional multiplication table restricted to entries at least 2 and products at most N : the set of tuples $(x_1, \dots, x_k)$ with every $x_i \geq 2$ and $x_1 x_2 \cdots x_k \leq N$. The starting observation is elementary and, we believe, a good pedagogical door into multiplicative number theory: the values manufactured by U_2 are exactly the composite numbers, and the primes are exactly the integers the table cannot make. (The table complement is a classical device — Sundaram’s sieve [11] generates the primes exactly this way — and the *de-duplicated* table, the set of distinct products, is the subject of the Erdős multiplication table problem [12, 13]; our concern below is with the table’s exact multiplicities rather than its cardinality.) An arithmetic amateur’s very natural program — *sum the integers up to N , subtract what the table makes, obtain the sum of the primes* — is therefore structurally sound, and fails only on multiplicity: the table ships duplicates, in quantities governed by the divisor function and, more fundamentally, by the number of prime factors $\Omega(m)$, which we call the *dimension* of m throughout.

This paper carries the program to completion and then generalizes it. The exact de-duplication weights turn out to be the alternating-harmonic sequence $1, -\frac{1}{2}, +\frac{1}{3}, \dots$ applied to the tower $U_1, U_2, U_3, \dots$; per integer this is Linnik’s identity [1, 2], the combinatorial expansion of $\log \zeta$. Around it we build a uniform, weighted, self-similar formula system with the following properties, each verified computationally at scales up to 10^9 : it computes $\sum_{A < p \leq B} p^j$ exactly for every $j \geq 0$ and every interval; it consumes no prime data at any point, its recursion bottoming out in empty tables; its interval structure is completely described by hyperbola shells, root bounds, and a finite quotient–root lattice; and it exhibits a *layer-birth law* organizing the dyadic scale 2^k as the birthday of dimension k .

We state our claims and non-claims plainly. The per-integer identity is Linnik’s; the $j = 0$ extraction line is Riemann’s Π -to- π inversion [3, 4] with Π supplied by table layers rather than an integral; and the resulting algorithms belong to the Legendre–Meissel combinatorial family [6, 7, 8, 9, 14], which contains asymptotically faster members. What we believe is worth recording: the derivation of the identity from the de-duplication ledger of a literal times table; the uniform treatment of counts, sums, and power sums as one three-line induction with a weight dial; the duplicated-mass cancellation theorem (Section 4); the interval taxonomy and layer-birth law (Section 5); the fully prime-free self-similar dust (Section 3); and the lattice-growth observable at prime steps (Section 6). Of these, the computational core — prime counts and prime power sums from Linnik’s identity, with no prime input — is anticipated by McKenzie [14], whose $x^{2/3}$ -class algorithm predates and asymptotically improves on our implementation; the remaining items we have not found presented in this form, though we regard that as a statement about our searches.

2 The table frame

Write $D_k^*(n)$ for the number of ordered factorizations of n into k factors, each at least 2; thus $D_1^*(n) = 1$ for $n \geq 2$, $D_2^*(n) = d(n) - 2$, and $D_k^*(n) = 0$ unless $\Omega(n) \geq k$, since dimension is additive and each factor carries dimension at least 1. Two appearance laws anchor intuition. In the (unrestricted) k -fold table the multiplicity of n is $d_k(n) = \prod_{p^a \parallel n} \binom{a+k-1}{k-1}$ — the Euler coefficients of $\zeta(s)^k$ [10] — so a squarefree n of dimension ω appears k^ω times: base the dimension of the space, exponent the dimension of the number. In the restricted table, primes appear only

in U_1 : a prime admits no factorization into two factors ≥ 2 . This is the survival mechanism of everything below.

The $x \leftrightarrow y$ fold of U_2 along its diagonal is Dirichlet's hyperbola device [5]; the diagonal itself carries the perfect squares, with the closed form $\sum_{k=2}^s k^2 = \frac{s(s+1)(2s+1)}{6} - 1$, $s = \lfloor \sqrt{N} \rfloor$, and after folding, the residual duplicates are exactly the values with several distinct two-part factorizations — necessarily of dimension ≥ 3 . (At $N = 100$: the folded interior square repeats only 12, 16, 18, 20, 24, 30, 36, 40, of dimensions 3, 4, 3, 3, 4, 3, 4, 4.)

3 The de-duplication identity and the formula system

Lemma 1 (Per-integer ledger; Linnik). *For every integer $n \geq 2$,*

$$\sum_{k \geq 1} \frac{(-1)^{k-1}}{k} D_k^*(n) = \frac{\Lambda(n)}{\log n} = \begin{cases} 1/j, & n = p^j, \\ 0, & \text{otherwise,} \end{cases}$$

the sum terminating at $k = \Omega(n)$.

The generating identity is $\log \zeta = \log(1 + F) = F - \frac{F^2}{2} + \dots$ with $F = \zeta - 1$; the ledger of each composite closes at its own dimension, and primes survive with weight 1 because they inhabit only U_1 . An equivalent integer-weight system uses dimension parity, $\mu(d) = (-1)^{\Omega(d)}$ on squarefree sieving patterns (Legendre's sieve); we work with the harmonic weights throughout.

Fix a weight $j \geq 0$ and define the layer totals $T_k^{(j)}(y) = \sum_{n \leq y} n^j D_k^*(n)$. The whole system is three lines:

$$\text{base: } T_1^{(j)}(y) = \sum_{n=2}^y n^j \quad (\text{Faulhaber closed form}), \quad (1)$$

$$\text{step: } T_k^{(j)}(y) = \sum_{d=2}^{\lfloor y/2^{k-1} \rfloor} d^j T_{k-1}^{(j)}(\lfloor y/d \rfloor), \quad (2)$$

$$\text{extraction: } S_j(y) = \sum_{k=1}^{\lfloor \log_2 y \rfloor} \frac{(-1)^{k-1}}{k} T_k^{(j)}(y) - \sum_{a=2}^{\lfloor \log_2 y \rfloor} \frac{1}{a} S_{aj}(\lfloor y^{1/a} \rfloor). \quad (3)$$

Theorem 1 (Prime power sums, prime-free). *For every $j \geq 0$ and $y \geq 1$, $S_j(y) = \sum_{p \leq y} p^j$; in particular $S_0 = \pi$ and S_1 is the prime-sum function, and for any interval, $\sum_{A < p \leq B} p^j = S_j(B) - S_j(A)$. The recursion (3) is self-similar (the dust calls the same S at root scales with multiplied weight) and terminates without any prime input: at $y \leq 3$ all layers with $k \geq 2$ vanish because the tables are empty, and the formula returns $\sum_{n \leq y} n^j$ over $\{2, 3\} \cap [1, y]$ of its own accord.*

The alternating combination in (3) is Riemann's prime-power counting function (weighted), and the extraction line is his Π -to- π inversion made recursive; the recursion depth is $O(\log \log y)$. As a worked instance, at $y = 30$, $j = 0$: the layers are 29, 52, 32, 5, giving $29 - \frac{52}{2} + \frac{32}{3} - \frac{5}{4} = \frac{149}{12} = \Pi(30)$; the dust is $\frac{\pi(5)}{2} + \frac{\pi(3)}{3} + \frac{\pi(2)}{4} = \frac{29}{12}$; and $\pi(30) = \frac{149}{12} - \frac{29}{12} = 10$. All rungs are linked by Abel summation, so counts rebuild sums and sums rebuild counts; and unit windows give membership, $S_1(n) - S_1(n-1) \in \{0, n\}$, so the aggregate formulas determine the set of primes itself by bisection.

4 The duplicated-mass theorem

Return to the amateur’s program in its corrected cage: the folded region $2 \leq x \leq \lfloor \sqrt{N} \rfloor$, $x \leq y$, $xy \leq N$, which manufactures every composite $\leq N$. Its raw sum is $R(N) = \frac{1}{2}(T_2^{(1)}(N) + Q(N))$ with Q the diagonal closed form above; the duplicated mass is $D(N) = R(N) - \sum_{m \leq N \text{ composite}} m$.

Theorem 2 (The 2D table cancels from its own duplication bill).

$$D(N) = \frac{Q(N)}{2} + \sum_{k \geq 3} \frac{(-1)^{k-1}}{k} T_k^{(1)}(N) - \sum_{p^j \leq N, j \geq 2} \frac{p^j}{j}.$$

The proof is three lines from Lemma 1: the composite indicator is $\frac{D_2^*}{2} - \frac{D_3^*}{3} + \dots$ up to prime-power dust, and upon subtracting, the $T_2^{(1)}$ terms annihilate. The duplication of the two-dimensional table is settled entirely by the squares and the tables of dimension three and higher — in agreement with the hand ledger, where every post-fold repeat offender had dimension ≥ 3 . Verified exactly: $D(100) = 4593$ from $192 + \frac{21426}{3} - \frac{13441}{4} + \frac{4112}{5} - \frac{640}{6} - \frac{5789}{60}$, and $D(1000) = 974,163$. The dust term is itself produced prime-free by Theorem 1.

5 Interval structure

Proposition 1 (Where the intervals live). *(i) For a target interval $(A, B]$ of values, every layer is summed over the hyperbola shell¹ $A < x_1 \cdots x_k \leq B$; the interval constrains the product, never the coordinates. (ii) In the sorted fold, the coordinates are root-bounded: $x_1 \leq B^{1/k}$ and inductively $x_i \leq (B/(x_1 \cdots x_{i-1}))^{1/(k-i+1)}$. (iii) The system (1)–(3) evaluates only at the closure of $\{B\}$ (and $\{A\}$) under $y \mapsto \lfloor y/d \rfloor$ and $y \mapsto \lfloor y^{1/a} \rfloor$ — a finite, self-similar lattice of $O(\sqrt{B})$ points (the quotient set alone has cardinality $2\sqrt{B} + O(1)$, exactly determined in [19]; e.g. 642 points for $B = 10^5$).*

Proposition 2 (Layer-birth law). $T_k^{(j)}(y) = 0$ for $y < 2^k$, and the contribution of layer k to its own dyadic window $(2^{k-1}, 2^k]$ is the single tuple $(2, 2, \dots, 2)$, of value 2^k and multiplicity one. Consequently the extraction truncates at $k = \lfloor \log_2 y \rfloor$, and the dyadic window of index k is the birth window of dimension k , containing exactly one newborn.

6 Implementation and verification

A reference implementation (`prime_sum_formula.py` in the code repository [23], about one hundred lines) realizes (1)–(3) with exact rational arithmetic, memoization over the quotient-root lattice, and hyperbola block-grouping of the step (2), so each lattice node costs $O(\sqrt{y})$; observed total cost scales near $y^{3/4}$. Every returned value is asserted to have denominator 1 before use — the harmonic fractions must annihilate, and in all runs they did.

Test suite. Forty interval tests, all exact against an independent sieve: the twenty dyadic windows to 2^{20} (sums); edge cases including the empty interval, the prime-free window $(113, 126]$,

¹“Shell” here means a product level region, the difference of two k -dimensional Dirichlet hyperbola regions — not the hyperbolic shells (level sets of indefinite quadratic forms) of the lattice-point literature.

a unit window, and a window containing 2^{10} to exercise the dust; eight random intervals below 10^6 ; five count tests (weight $j = 0$), including $\pi(10^6) = 78,498$; and the spans $(10^6, 10^7]$ and $(1, 10^7]$, the latter returning 3,203,324,994,356.

Billion-scale validation. With published values as the oracle ($\pi(10^9) = 50,847,534$; 999,999,937 the largest prime below 10^9), the implementation computed $\pi(999,999,936) = 50,847,533$ in 176 s, $\pi(999,999,937) = 50,847,534$ in 1 s, and $\pi(10^9) = 50,847,534$ in 18 s, touching 63,719 lattice points in total: the formula’s verdicts — that 999,999,937 is prime, is prime number 50,847,534, and is the last prime below one billion — agree with the record values, with no sieve and no primality test in the computation.

Remark 1 (Primality as lattice growth). Since $\lfloor (y+1)/d \rfloor > \lfloor y/d \rfloor$ exactly when $d \mid y+1$, the evaluation lattice of $y+1$ exceeds that of y only at divisors of $y+1$. A prime step therefore adds a single lattice point — observed above as the $176\text{ s} \rightarrow 1\text{ s}$ collapse and a lattice increment of exactly one. The system’s cost profile feels primality before its arithmetic announces it. The underlying object has a literature of its own: the floor-quotient set $\{\lfloor y/d \rfloor\}$ carries a partial-order structure studied by Lagarias and Richman [18], its cardinality is exactly determined [19], and the primes *inside* it are actively studied [20, 21]. The observable here is of a different kind: the bare cardinality of the quotient set grows at every step and does not discriminate primes, whereas the growth of the closed evaluation lattice is supported exactly on the divisors of $y+1$ — kin to the folklore reading of $D(n) - D(n-1) = d(n)$ for the divisor summatory function, minimal exactly at the primes.

7 Related work, limitations, open doors

Lemma 1 is Linnik’s identity [1], standard in sieve theory [2] and developed into $x^{2/3}$ -class algorithms for $\pi(x)$ and prime power sums by McKenzie [14] (unrefereed manuscripts, whose complexity analysis is McKenzie’s own); within the same lineage, Sladkey’s $O(x^{1/3})$ divisor-summatory algorithm [15] is the natural subroutine for pressing below the $x^{2/3}$ class. The $j = 0$ extraction is Riemann’s inversion [3, 4]; Legendre’s sieve is the dimension-parity twin; the combinatorial prime-counting lineage [6, 7] and its prime-power-sum extensions [8, 9, 17] contain algorithms of better exponent ($x^{2/3}$ -class) than our $x^{3/4}$ -class implementation, with record prime-sum computations to 10^{26} [17]; and an elementary route of Hirsch, Kessler, and Mendlovic reaches $\tilde{O}(\sqrt{x})$ for counts and sums alike [16]. The companion paper [22] develops a different, certified route to interval prime sums via Chebyshev-weighted generalized factorials. The present system computes any functional of primes defined by *individual* membership — intervals here; residue classes are an open door, since the ledger survives residue-restricted weights and the Faulhaber corrections remain closed over progressions — but no weighting can express *joint* conditions across several integers (twin primes, gaps): the per-integer collapse cannot see correlations, which is the field’s frontier, not this construction’s defect. Finally, the system is a formula *family* of $\sqrt{N}$ -length evaluations, not a fixed-length closed form; the obstruction to further compression is the standard one, that the exact fluctuations carry the zeta spectrum.

Disclosure of AI use

In accordance with publisher policies on the use of artificial intelligence, the author discloses the following. This paper was developed in an extensive collaboration with Claude (Anthropic;

Claude Fable 5, accessed through the claude.ai interface, 2026). The direction of the work — the research programme of exact moment systems for interval primes — together with all decisions about scope, claims, and submission, are the author’s. Within that direction, the AI system contributed substantially to the mathematical development and proofs, wrote the verification code, drafted and revised the manuscript text, and assisted with literature search; the reason for its use was to enable a single non-specialist author to develop, check, and document the material to a publishable standard. Every numerical and mathematical claim was subsequently machine-verified by the test suite and billion-scale validation reported in Section 6, with the verification code openly available in the cited repository [23]. The author has reviewed the entire manuscript and takes full responsibility for its content, including the accuracy of the mathematics and of all references.

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